

TLS1485, 1585, TLS463 System Install/Calibration Checklists

January 11, 2023

Purpose

Use this checklist when ~~performing the installation~~installing in accordance with following the TI Manuals. This list is not intended to replace the TI manual procedures.

Instructions

Choose one or both options

- 1) Print hard copy, then use during deployment and sign-off in the field, return completed copy for the scan to archive
- ~~2)~~ Print a hard copy and laminate, ~~use;~~ use a grease pen to annotate in the field

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Step 1: Pre-deployment Checklist

- ☐ Site plan supports the desired approach
- ☐ Site plan applies to correct runway end
- ☐ Site plan complies with siting criteria
- ☐ Site plan complies with critical-area criteria
- ☐ Frequency transmit authorization has been coordinated satisfactorily
- ☐ Date and time for calibration aircraft is scheduled
- ☐ System transport logistics from factory to airport have been determined
- ☐ System transport at airport to operating location next to runway have been determined
- ☐ ATC Console map display features have been determined

Location _____

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Step 2: ASA Checklist

- ☐ Quadpod installed at location according to site plan
- ☐ High-res antenna is between Ref antenna and ESA
- ☐ Low-res antenna cycle is installed as intended, 20"/30"
- ☐ Beam is within 10 degrees of perpendicular to runway
- ☐ Beam has been leveled
- ☐ ASA/ESA electronics selector switch set to ASA
- ☐ Coax cables labels match the attachment point label
- ☐ Cables are properly routed and secure
- ☐ Power/Fiber Cable seated properly; strain-relief attached
- ☐ ASA is anchored securely to the ground using augers and turnbuckles

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Step 3: ATA Checklist

- ☐ Tripod installed at location according to site plan
- ☐ Coax cable labels match the connector labels
- ☐ ATA "signal out" cable connected to the MED channel on the ASA AOA Sensor
- ☐ Coax cable seated properly and strain relief attached
- ☐ Anchor the ATA using the augers and turnbuckles

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Step 4: Cal/BIT Checklist

- ☐ Tripod installed at location according to site plan
- ☐ Main beam of antenna is pointed toward the ESA
- ☐ Power/Fiber Cable seated properly; strain-relief attached
- ☐ Attenuate Cal/BIT output to be within each sensors power range
- ☐ Cal/BIT anchored using the augers and turnbuckles

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Step 5: Interrogator Checklist

- ☐ Tripod installed at location according to site plan
- ☐ The suppression antenna is pointed in the direction opposite the guidance volume
- ☐ Coax cable labels match the connector labels
- ☐ Anchor the tripod using the augers and turnbuckles

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Step 6: ESA/Uplink Pre-Stand Checklist

- ☐ Base plate location per site plan
- ☐ Base installed within 10 degrees of ESA
- ☐ Augers installed in the correct locations [21' for TOA on top or 25' for 30 foot tower from base](#)
- ☐ Four out of six guy-wires attached to the correct connection points on towers and at augers
- ☐ Turnbuckles expanded to within the last inch of thread
- ☐ Uplink antennas installed to point at the approach
- ☐ ESA antennas are installed from bottom to top: Ref, Low, Med, and High
- ☐ ESA directional antennas are pointed toward the approach
- ☐ Coax cable labels match the connector labels
- ☐ Obstruction light cable is attached
- ☐ All cables are secured with Velcro straps to ~~the a~~ tower [leg](#)
- ☐ [Antenna separations are measured and recorded for later entry into the MIU survey screen](#)
- ☐ Gin pole installed correctly/prepped for lift
- ☐ Winch is installed and cable is evenly wound on drum

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Step 7: ESA/Uplink Post-Stand Checklist

- ☐ ESA tower is plumb, was scanned with theodolite
- ☐ Tilt sensor cable labels match connector labels
- ☐ Guy wires are properly tensioned upper 600lbs / lower 400 lbs
- ☐ Coax cable labels match the connector label for all cable connections
- ☐ ASA/ESA electronics selector switch set to ESA
- ☐ All coax is secured with Velcro straps
- ☐ Uplink coax cable labels match the connector at the antenna and shelter panel
- ☐ Power/Fiber Cable seated properly; strain-relief attached

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Step 8: Post-Survey

- ☐ Prism pole adjusted to theodolite eyepiece height
- ☐ FTM Theodolite position surveyed and corresponds to the approach threshold crossing height
- ☐ All structures are anchored properly
- ☐ [Verify power plugs did not detach during shipment, reattach where necessary](#)
- ☐ Archive previous data
- ☐ Elevation recorded correctly (+/- sign)
- ☐ Survey data entered into the system (rack A/B)
- ☐ ASA low-res spacing entered as intended 20" or 30" to match actual antenna spacing
- ☐ [ESA antenna separations are entered](#)
- ☐ Verify all CAL/BIT received amplitudes at the ASA are within 50 millivolts
- ☐ Verify TOA jitter on all channels is less than 2.8 ns
- ☐ Verify all CAL/BIT received amplitudes at the ESA are about the same level
- ☐ Survey distances in units of feet, enter survey distances in units of feet
- ☐ Survey data verified by a second person
- ☐ [BIT/RAM is enabled](#)
- ☐ [Guidance RAM is enabled](#)
- ☐ Tune the F1 delay to have jitter less than 3 ns for the ATA, ESA, and ASA
- ☐ BIT stable with 10 cycles
- ☐ Monitor limits set for ATA, ASA, ESA, Uplink 1-4, and Int
- ☐ TOA calibrated using SysCal tool, entered on MI

Location _____

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Step 9: Pre-Calibration Checklist

- ☐ Theodolite batteries charged
- ☐ Hand-held radios charged
- ☐ Theodolite can be test pinged before briefing pilot
- ☐ Set the RCU to be used with FTM on the MIU settings screen
- ☐ Pilot briefed on the required approaches.
- ☐ Pilot knows to ask for a transponder code assignment
- ☐ Pilot knows what ILS frequency to tune
- ☐ Pilot knows to turn on landing lights
- ☐ Pilot knows the first approach is visual, cannot trust Loc
- ☐ Approach.dat has an offset option if not using the centerline
- ☐ Pilot briefed on offset theodolite approach if used
- ☐ Pilot briefed on verification low approach only if theodolite is on the runway

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Step 10: Theodolite Setup Checklist

- ☐ Theodolite properly leveled at surveyed location
- ☐ FTM Cable correctly routed with strain relief
- ☐ Verify pings from FTM software
- ☐ Theodolite azimuth zeroed on the back-site/centerline (the direction that the planes approach from/positive X)
- ☐ Theodolite knobs adjusted to allow access to service volume elevation 0 to 7 degrees, azimuth +/-3 degrees
- ☐ Handheld ground-to-air radio is tuned to the correct frequency

Location _____

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Step 11: Use this real-time Calibration Checklist

This checklist is intended to augment the procedure described in the manual 020-00071

- ☐ Theodolite is in position and level
- ☐ Handheld comms with theodolite operator is good to go
- ☐ Select the FTM tool from the MIU Tool menu, press “L,” and then “Ok.”
- ☐ Select Test Mode from the MIU System menu option
- ☐ Enter the transponder code on the MIU test mode screen
- ☐ Select the desired approach profile and GTU from the drop-down menus
- ☐ Press “Acquire” when the pilot reports turning inbound
- ☐ Keep the theodolite operator updated on the aircraft position
- ☐ After the theodolite operator reports he is tracking, press record on the FTM tool
- ☐ At the end of the approach, press stop on the FTM tool and save the file
- ☐ Follow the Syscal SW tool instructions in 020-00071 to calibrate the TLS
- ☐ Save the new cal.dat to the MIU, Base1, and Base2
- ☐ OPTION: Use the Mode-C calibration option if theodolite data is missing or challenged by visibility
- ☐ 1st level approach successful
- ☐ 2nd level approach successful
- ☐ Diff files generated and saved in the correct location
- ☐ Syscal tool used the correct Glideslope offset for low res and high (i.e., 1.0 or, 4.6 degrees for example)
- ☐ Syscal plots were normal and saved
- ☐ Cal.dat file saved correctly to MIU, Base1, and Base2
- ☐ Verification approach completed and signal verified by the pilot

Location _____

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Step 12: Post-Calibration Checklist

- ☐ Enable guidance ram files
- ☐ Tune Diversity Algorithm
- ☐ Disable ~~bitram~~-BITRAM files
- ☐ Set localizer volume to 25 nm and +/- 42 degrees
- ☐ Set glide slope volume to 12 nm +/- 12 degrees
- ☐ Put away all tools used for installation/calibration in their correct locations
- ☐ RCU/Ubiquiti link is installed
- ☐ Breaker on the generator is set to *ON*
- ☐ Generator auto switch-over tested
- ☐ Ob-lights are all on; enabled on HTML page
- ☐ Wrap all coax connections with self-vulcanizing tape
- ☐ Select preference for "All GTUs" on RCU
- ☐ Perform Site Acceptance Test (SAT)

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Step 13: IAP Checklist

- ☐ GTU(s) is programmed to the correct frequency
- ☐ Glide slope angle is set to the desired glide slope with correct TCH
- ☐ Audio markers on Localizer are set to the correct distances

Location _____

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